CT-1 Crash Sheet

Introduction to Probability and Statistics (CSE/STAT 251)

Syllabus basis: class notes Stat-01 – Stat-07 • First-priority pattern: "32" CT-1 paper • Full marks pattern: 30

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1 Priority chart — study in this order

#	Topic	Evidence (past papers)	Expected weight
1	Box plot, five-number summary, IQR outlier rule	32-CT Q1 (10m); 2021 Q2	$\approx 10/30$
2	Histogram: Sturges' rule + bin width + drawing	32-CT Q4 (6m); notes	$\approx 6/30$
3	Classification vs. regression; target/features; data types	32-CT Q5 (6m)	$\approx 6/30$
4	Percentiles → fences (abnormal high/low rule)	32-CT Q2 (3m)	$\approx 3/30$
5	Mean vs. median, skewness, which to report	32-CT Q3 (3m)	$\approx 3/30$
6	Definitions: outlier, population, sample	32-CT Q6 (2m)	$\approx 2/30$
7	3σ / z-score rule, tolerance levels	notes (heavily)	backup
8	Variance, SD, comparing two datasets	notes worked ex.	backup
9	Covariance & correlation	2021 Q3(c)	backup
10	ML families, regression cost functions	notes Stat-07	short theory

Question-pattern map (32 CT): Q1 → §7; Q2 → §6+§7; Q3 → §3; Q4 → §8; Q5 → §10; Q6 → §2. All six solved in §12.

How to use: Pass 1 — read §2–§11 (≈2.5 h). Pass 2 — solve §12 without looking (≈1 h). Pass 3 — the last-page formula sheet, 30 min before the hall.

2 Foundations (definitions you must write verbatim)

Direct 2-mark answers (32-CT Q6 style)

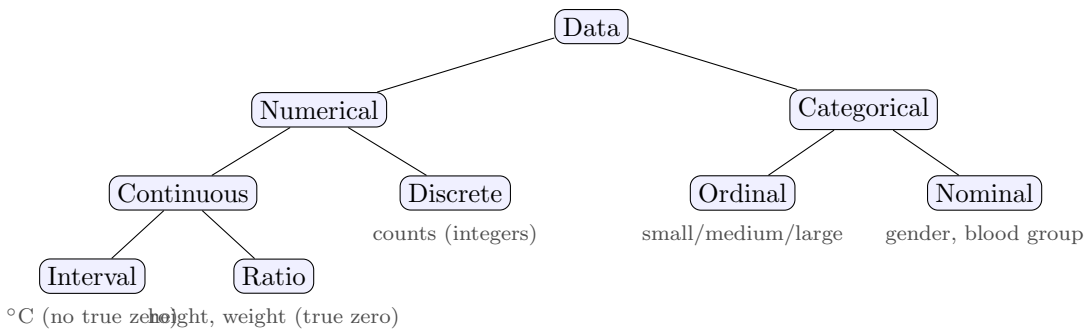
Outlier: a value abnormally far from the rest of the data; formally, x is an outlier if $x < Q_1 - 1.5\,\text{IQR}$ or $x > Q_3 + 1.5\,\text{IQR}$ (equivalently outside $\bar{x} \pm 3s$).
Population: the *entire* group of individuals/items under study (the sector where the research happens). Size $= N$, mean $= \mu$. Ex: all students of a university.
Sample: a small *subset* of the population selected for analysis. Size $= n$, mean $= \bar{x}$.

Data vs. information: data = raw recorded facts; information = processed/organized data with meaning.

Dataset = table. X = input columns (features / independent variables / predictors), Y = output column (target). n = rows (samples), m = columns (features), shape $= (n, m)$. One sample = one row = one data point = one *feature vector* = a point in m-dimensional space.

Structured data: representable in rows/columns (databases) — easy to analyze. **Unstructured:** images, video, mails (files/media).

2.1 Types of data



Categorical data = data grouped into categories/labels rather than measured numerically; classifies items by characteristics.

2.2 Descriptive vs. inferential statistics

Descriptive	Inferential
Collects, organizes, summarizes, *describes* a dataset. No prediction/decision.	Uses a *sample* to make conclusions, predictions or decisions about the whole *population*.
Tools: mean, median, mode, range, variance, IQR, outliers, plots.	Tools: estimation, hypothesis testing.

Why statistics? Backbone of Machine Learning, Data Science, Deep Learning (a dataset is what an ML model *needs*). **Workflow:** data collection → arrange/represent (graphical, EDA) → analysis → interpretation.

EDA (Exploratory Data Analysis): the process of examining, summarizing and visualizing data to understand its patterns, detect outliers, and identify relationships *before* further analysis.

Univariate analysis = analyzing one column at a time. It helps to: understand the distribution of a single variable, find the central value, detect missing values, detect outliers, and replace missing values with an appropriate (central/expected) value. Teacher's convention: mean → 1st moment M_1, median → 2nd moment M_2, mode → 3rd moment M_3.

3 Measures of central tendency

Which one? (memorize this table)

Situation	Center	Spread
No outlier	**Mean**	Range
Outlier present	**Median**	IQR
Categorical data	**Mode**	—

3.1 Mean

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n} = \frac{1}{n}\sum_{i=1}^{n} x_i \qquad \text{(sample: } \bar{x}\text{, population: } \mu\text{)}$$

Class example — outlier destroys the mean

Ages $x = \{24, 25, 26, 23, 24, 22, 27, 24, 60\}$, $n = 9$ (60 = outlier). $\bar{x} = \frac{24 + 25 + \cdots + 60}{9} = \frac{255}{9} = 28.33 \approx 28$ years. Without 60, the center is ≈ 24. So 28 is *misleading*: when an outlier exists, $\boxed{\text{average value} \neq \text{center value}}$ ⇒ use the **median**.

3.2 Median (algorithm — write these steps in exam)

1. Index all values; **sort ascending**.
2. $mid = \frac{start + end}{2} = \frac{n+1}{2} = 50\%$ of $(n+1)$.
3. n odd: median $= x_{mid}$. n even: median $= \frac{x_{mid} + x_{mid+1}}{2}$ (interpolation).

Median of the class dataset

Sorted: $22, 23, 24, 24, \underline{24}, 25, 26, 27, 60$; $n = 9$; $mid = \frac{1+9}{2} = 5$ ⇒ median $= x_5 =$ **24** years. 50% of data lies left of the median, 50% right.

3.3 Mode and data distribution

Data distribution $f(x)$: table of each unique value x and its frequency $f(x)$. **Mode** = most frequent data value (frequency must be > 1). Unimodal = one mode; bimodal = two; multimodal = more than two; **no-modal** dataset = every value has the same frequency $\Rightarrow$ no mode.

Grades (categorical $\Rightarrow$ mode)

grades: A, A^+, A, A, A, B, B^- $\Rightarrow f(\text{A}) = 3$ (highest) $\Rightarrow$ mode = A (unimodal).

3.4 Mean vs. median $\Rightarrow$ skewness (needed for 32-CT Q3)

Observation	Shape	Cause
mean $>$ median	**right / positively skewed**	few very large values pull mean up
mean $<$ median	left / negatively skewed	few very small values pull mean down
mean $\approx$ median	symmetric	no strong outliers

Mean and Range are *affected* by outliers; Median and IQR are *robust* (not affected much).

4 Measures of spread (dispersion): R, s^2, s

Variation of one point: $x_i - \bar{x}$. Total variation $= \sum_{i=1}^{n} |x_i - \bar{x}|$. Mean variation $= \dfrac{1}{n-1} \sum_{i=1}^{n} |x_i - \bar{x}|$ (the $n - 1 =$ **DOF**, degrees of freedom).

Sample variance $s^2 = \dfrac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2$ **Population variance** $\sigma^2 = \dfrac{1}{N} \sum_{i=1}^{N} (x_i - \mu)^2$

Standard deviation $s = \sqrt{s^2}$ Ex: $s^2 = 4$ years^2 $\Rightarrow s = 2$ years **Range** $R = \max - \min$

Comparing two datasets (straight from notes — likely CT question)

DS-1 $= \{78, 79, 80, 81, 82\}$, DS-2 $= \{60, 70, 80, 90, 100\}$.

Dataset	$\bar{x}$	min	max	R	s^2	s
DS-1	80	78	82	4	2.5	1.58
DS-2	80	60	100	40	250.0	15.81

Interpretation: same mean, but DS-1 samples are *tightly concentrated* around $\bar{x}_{DS1}$, while DS-2 samples are far more *spread out / dispersed* from $\bar{x}_{DS2}$. R, s^2, s are the measures of spread-out-ness.

4.1 Tolerance level

Tolerance = maximum acceptable amount of error/deviation. Levels: $\bar{x} \pm ns$.

$$20 \pm s = [18, 22], \qquad 20 \pm 2s = [16, 24], \qquad \bar{x} \pm 3s = [14, 26]$$

$[\bar{x} - 3s,\ \bar{x} + 3s]$ is the **tolerance range / standard**: values inside are acceptable (normal).

5 Outlier detection method 1: the 3σ / z-score rule

Rule (write both lines)

x is an **outlier (abnormal)** if $x < \bar{x} - 3s$ or $x > \bar{x} + 3s$.
x is **normal** if $\bar{x} - 3s \le x \le \bar{x} + 3s$. Equivalently $z = \dfrac{x - \bar{x}}{s}$; outlier if $|z| > 3$.

Class example

$\bar{x} = 20$, $s = 2 \Rightarrow$ normal range $[20 - 6,\ 20 + 6] = [14, 26]$. $x = 18 \Rightarrow$ normal; $x = 30 \Rightarrow$ outlier.

Outlier detection methods (name both in theory answers): (i) z-score (3σ) method, (ii) Boxplot (IQR) method.

6 Quartiles & percentiles

Quartile system divides sorted data into **4** equal parts (25% each): Q_1 = 25th percentile, Q_2 = median = 50th, Q_3 = 75th. **Percentile system** divides into **100** parts. P_k = the value below which (or equal to which) k% of the data lies.

$$\boxed{\text{IQR} = Q_3 - Q_1} \qquad \text{(middle 50\% of the data lies between } Q_1 \text{ and } Q_3\text{)}$$

6.1 Finding P_k: the rank method (teacher's algorithm)

`get_kth_percentile`(x, n, k):

1. Sort ascending. Compute rank/position $r = \dfrac{k}{100}(n + 1) = k\%$ of $(n + 1)$.
2. If r is an integer: $P_k = x_r$. Else, with $d = r - \lfloor r \rfloor$:

Case	Formula	Notes
Lower interpolation	$P_k = x_{\lfloor r \rfloor}$	floor
Higher interpolation	$P_k = x_{\lceil r \rceil}$	ceiling
Mid-point interpolation	$P_k = \dfrac{x_{\lfloor r \rfloor} + x_{\lceil r \rceil}}{2}$	average
Linear interpolation	$P_k = x_{\lfloor r \rfloor} + d\left(x_{\lfloor r \rfloor + 1} - x_{\lfloor r \rfloor}\right)$	**most preferred** $\star$

Type-1

$x = \{5, 7, 10, 12, 14, 15, 18, 21, 25, 30, 35\}$ sorted, $n = 11$.
P_{50}**:** $r = 50\% \times 12 = 6$ (integer) $\Rightarrow P_{50} = x_6 =$ **15**.
P_{30}**:** $r = 30\% \times 12 = 3.6$. Lower: $x_3 = 10$; higher: $x_4 = 12$; midpoint: 11; **linear**: $10 + 0.6(12 - 10) =$ **11.2**.
P_{75}**:** $r = 75\% \times 12 = 9 \Rightarrow P_{75} = x_9 =$ **25**.

Second drill (n

$x = \{5, 10, 12, 14, 15, 18, 20, 25, 30, 35\}$. P_{50}: $r = \frac{50}{100}(11) = 5.5 \Rightarrow 15 + 0.5(18 - 15) =$ **16.5**. P_{30}: $r = 3.3 \Rightarrow 12 + 0.3(14 - 12) =$ **12.6**.

6.2 Type-2 (reverse): what percentile is a given value?

$$k = \frac{r}{n} \times 100, \qquad r = \text{number of values} \le v \text{ (its sorted position)}$$

From notes

In $\{22, 23, 24, 24, 24, 25, 26, 27, 60\}$, the value 24: five values are ≤ 24, so $k = \frac{5}{9} \times 100 = \mathbf{55.6}\%$.

Exam trap

rank $\neq$ percentile rank. Rank r = position in sorted data. Percentile rank k = the percentage. Never mix the two symbols.

7 Outlier detection method 2: IQR rule + Box plot

Five-number summary: (1) Minimum, (2) Q_1, (3) Q_2 (median), (4) Q_3, (5) Maximum. A **box plot** is a graphical representation of the dataset using the five-number summary. Min and Max are *influenced by outliers*; Q_1, Q_2, Q_3 are robust.

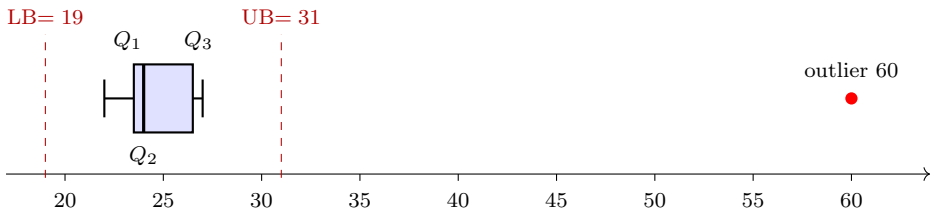
IQR fences (empirical formula)

$$\text{Lower bound} = Q_1 - 1.5 \times \text{IQR} \qquad \text{Upper bound} = Q_3 + 1.5 \times \text{IQR}$$

Decision rule: LB $\leq x \leq$ UB $\Rightarrow x$ normal; $x <$ LB or $x >$ UB $\Rightarrow x$ **outlier**.
This is an *empirical* formula (based on observation/experiment, cannot be mathematically proved), unlike *theoretical* ones such as $(a+b)^2 = a^2 + 2ab + b^2$.
Extension (2021-exam style): *mild* outlier = beyond 1.5×IQR fences but within 3×IQR fences; *extreme* outlier = beyond $Q_1 - 3\,\text{IQR}$ or $Q_3 + 3\,\text{IQR}$.

Full worked example (the class dataset) — reproduce this pattern in exam

$x = \{24, 25, 26, 23, 24, 22, 27, 24, 60\}$, $n = 9$.
P-1 Sort: $22, 23, 24, 24, 24, 25, 26, 27, 60$. **P-2** $Q_1 = 23.5$, $Q_2 = 24$, $Q_3 = 26.5$ (rank method: $r_{Q_1} = 25\% \times 10 = 2.5 \Rightarrow 23 + 0.5(1) = 23.5$; $r_{Q_3} = 7.5 \Rightarrow 26 + 0.5(1) = 26.5$).
P-3 IQR $= 26.5 - 23.5 = 3$. **P-4** LB $= 23.5 - 1.5(3) =$ **19**; UB $= 26.5 + 1.5(3) =$ **31**.
P-5 $60 > 31 \Rightarrow 60$ is an **outlier**. All others normal.
P-6 Box plot: box 23.5 → 26.5, line at 24; whiskers to min = 22 and to the *largest value within the upper bound* = 27; plot 60 as a dot.



$Q_1 = 23.5$, $Q_2 = 24$, $Q_3 = 26.5$; whiskers at 22 and 27 (largest value within UB).

Reading skew from a box: median closer to Q_1 (upper box-half longer) $\Rightarrow$ positively (right) skewed; closer to $Q_3 \Rightarrow$ negatively skewed.

8 Graphical representation & histogram construction

Six standard plots: (i) Bar chart, (ii) Histogram, (iii) Pie chart, (iv) Box plot, (v) KDE plot, (vi) Q–Q plot.

Bar chart	Histogram
For *categorical / discrete* data	For *continuous numerical* (floating-point) data
Bars have gaps; order of bars can be changed	No gaps; bins are ordered intervals
x-axis: categories, y-axis: frequency; compares categories	x-axis: bins, height = frequency of the bin

Histogram algorithm (Sturges' rule) — exactly as tested in 32-CT Q4

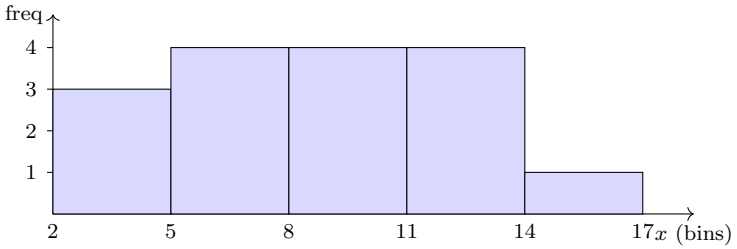
1. min, max, Range $R = \max - \min$, sample size n.
2. Number of bins $k = \lceil \log_2 n + 1 \rceil$ (Sturges' formula; $\lceil \cdot \rceil$ = round *up*).
3. Bin width $w = \left\lceil \frac{R}{k} \right\rceil$.
4. Bins are half-open $[a,\ a{+}w)$: **include** left edge, **exclude** right edge. Start at min.
5. Tally frequencies; draw bars with height = frequency (no gaps).

Quick $\log_2$ values: $\log_2 8 = 3$, $\log_2 15 \approx 3.91$, $\log_2 16 = 4$, $\log_2 32 = 5$, $\log_2 64 = 6$.

Class example

$x = \{2, 4, 4, 5, 6, 7, 7, 8, 9, 10, 10, 11, 12, 13, 13, 15\}$: $R = 15 - 2 = 13$; $k = \lceil \log_2 16 + 1 \rceil = \lceil 5 \rceil = 5$; $w = \lceil 13/5 \rceil = \lceil 2.6 \rceil = 3$.

Bin	Interval	Freq
B_1	$[2, 5)$	3
B_2	$[5, 8)$	4
B_3	$[8, 11)$	4
B_4	$[11, 14)$	4
B_5	$[14, 17)$	1



Exam trap

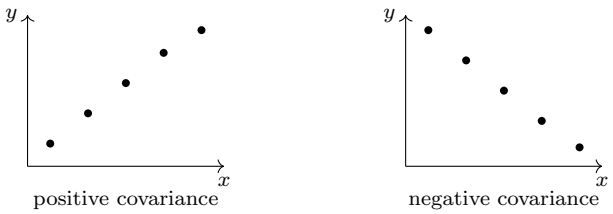
If $w = \lceil R/k \rceil$ comes out *exact* (no rounding up), the maximum lands on the last right edge and would be excluded. Fix: close the last bin, e.g. $[267, 295]$. If w was rounded up (like $2.6 \to 3$), the last bin already extends past max — no problem.

9 Covariance & correlation

Covariance $\mathrm{Cov}(x, y)$ explains the relationship between two variables x and y:

$$\text{sample: } \mathrm{Cov}(x,y) = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}) \qquad \text{population: } \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu_x)(y_i - \mu_y)$$

+ve Cov: $x \uparrow$ $y \uparrow$ and $x \downarrow$ $y \downarrow$ (same direction). −ve Cov: $x \uparrow$ $y \downarrow$ (opposite). Cov = 0: no *linear* relationship. Range $(-\infty, +\infty)$; carries units (e.g. hrs×marks) — so it cannot be compared across scales.



Correlation = normalized covariance ⇒ unitless, scale-free:

$$r = \frac{\mathrm{Cov}(x,y)}{s_x\, s_y} = \frac{\sum(x_i-\bar{x})(y_i-\bar{y})}{\sqrt{\sum(x_i-\bar{x})^2}\sqrt{\sum(y_i-\bar{y})^2}}, \qquad r \in [-1,1]$$

Geometric meaning: $r = \cos\theta$ between the mean-centered vectors $\vec{X} = [x_i - \bar{x}]$ and $\vec{Y} = [y_i - \bar{y}]$ (from $\vec{X}\cdot\vec{Y} = \|\vec{X}\|\|\vec{Y}\|\cos\theta$). Also called **Pearson correlation coefficient**.

r	relation	r	relation
1	perfect +ve	−0.1 to −0.3	weak −ve
0.7–0.9	strong +ve	−0.4 to −0.6	moderate −ve
0.4–0.6	moderate +ve	−0.7 to −0.9	strong −ve
0.1–0.3	weak +ve	−1	perfect −ve
0	no linear relation		

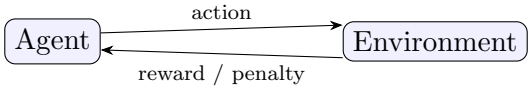
Worked (class data)

x(ToS hrs) $= \{2,4,6,8,10\}$, y(Marks) $= \{65,70,75,80,85\}$; $\bar{x} = 6,\ \bar{y} = 75$.
$\sum(x_i-\bar{x})(y_i-\bar{y}) = (-4)(-10)+(-2)(-5)+0+2(5)+4(10) = 100 \Rightarrow \mathrm{Cov} = 100/4 = \mathbf{25}$.
$s_x = \sqrt{40/4} = 3.16$, $s_y = \sqrt{250/4} = 7.91 \Rightarrow r = \dfrac{25}{3.16\times 7.91} = \mathbf{1.0}$: perfect positive linear relation (indeed $y = 2.5x + 60$).

10 Statistics → Machine Learning (32-CT Q5 territory)

Dataset $\xrightarrow{\text{need}}$ ML model → patterns. 1 time-period = 1 **epoch**.

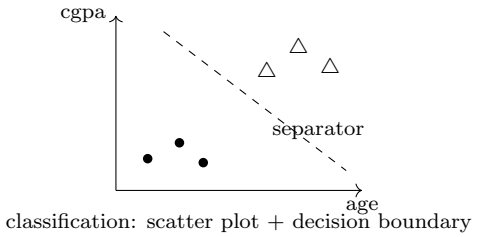
- **Supervised ML:** model trained on a *labelled* dataset (input x *and* output y given).
- **Unsupervised ML:** *unlabelled* dataset; finds structure. *Cluster* = group of objects (like batch = group of students).
- **Reinforcement learning:** an *agent* interacts with an *environment* by *actions* and learns from *rewards/penalties*.



Two types of supervised ML problems — the exam decision

Regression: output column is *continuous numerical* (floating-point) values. Ex: predict marks, predict salary.
Classification: output is *categorical* (labels: Pass/Fail, Yes/No, 0/1). The algorithm separates classes with a **decision boundary**: a separating line (2D), a plane (3D), a **hyperplane** (≥4D).

Space	Decision boundary	Dim.
2D	straight line	1D
3D	plane	2D
mD	hyperplane	$(m-1)$D



classification: scatter plot + decision boundary

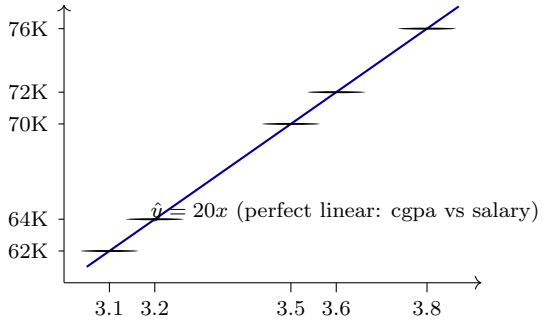
Q5-style drill (always answer in this order): (1) find the target column → (2) target type (continuous? categorical?) → (3) regression or classification → (4) list features, mark each numeric / categorical.

Regression analysis family: (i) Simple linear (one input column), (ii) Multiple linear (more than one input column), (iii) Polynomial, (iv) Logistic, (v) Ridge–Lasso.

11 Simple linear regression & cost functions

Model: $\hat{y} = wx + b$ (w = slope, b = intercept, $\hat{y}$ = *predicted* output, y = actual output). Best-fit regression line = the line representing the dataset in the best way.

Three dataset cases: **Case 1** perfect linear (all points on one line); **Case 2** kind-of linear (many candidate lines LR1, LR2, LR3 ⇒ pick minimum-cost one); **Case 3** non-linear (a line cannot fit; need polynomial etc.).



Error of one point (residual): $e_i = y_i - \hat{y}_i$. Total error/loss $= e_1 + e_2 + \cdots + e_n$.

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \qquad \text{SSE} = \text{RSS} = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

$$\text{MSE (mean square sum, MSS)} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2 \qquad \text{RMSE (RMSS)} = \sqrt{\text{MSE}}$$

The optimization problem (write this sentence)

Cost is a function of the predicted output: $C(\hat{y}_i)$. With $\hat{y}_i = wx_i + b$,

$$C(w,b) = \frac{1}{N}\sum_{i=1}^{N}\big(y_i - wx_i - b\big)^2.$$

Goal: find the w and b for which $C(w,b)$ is minimum — that line is the best-fit regression line (minimum error / minimum cost function).

Bonus (needed for 2021-style "fit a regression line"): the minimizing solution is

$$w = \frac{S_{xy}}{S_{xx}} = \frac{\sum x_i y_i - \frac{(\sum x_i)(\sum y_i)}{n}}{\sum x_i^2 - \frac{(\sum x_i)^2}{n}}, \qquad b = \bar{y} - w\,\bar{x}.$$

12 SOLVED: "32" CT-1 paper (first priority)

12.1 Q1 — Box plots of 4 search algorithms [FROM SYLLABUS]

Given five-number summaries (ms). First compute IQR and Range:

Algo	Q_1	Median	Q_3	Min	Max	IQR	Range
LS	40	55	70	30	200	30	170
BS	10	14	18	8	22	8	14
JS	18	24	30	15	36	12	21
ES	8	10	12	7	14	**4**	**7**

(a) Most consistent (stable): ES — it has the smallest IQR ($12-8=4$ ms) and the smallest range ($14-7=7$ ms), i.e. the least spread/dispersion of execution times, so its performance varies the least.

(b) Best for real-time (low latency + stable): ES — lowest median (10 ms < BS's 14 ms) *and* the tightest spread (IQR 4, range 7), so it is both fast and predictable, exactly what real-time systems need.

(c) Outlier in LS: IQR $= 70-40 = 30$. LB $= 40-1.5(30) = -5$; UB $= 70+1.5(30) = 115$. Max $= 200 > 115 \Rightarrow$ **200 is an outlier** (Min $= 30 > -5$, so no lower outlier).
Interpretation: LS usually runs in 40–70 ms, but occasionally hits its worst case ($\approx$ 200 ms, e.g. the key is at the end of a large list). Its tail latency is unpredictable, the mean is inflated by such runs, so LS is unreliable for time-critical applications.

12.2 Q2 — Abnormal heart-rate rule from percentiles [FROM SYLLABUS]

$P_{25} = 120$, $P_{50} = 150$, $P_{75} = 170$ bpm. IQR $= P_{75} - P_{25} = 50$.

$$\text{Abnormally low: } x < P_{25} - 1.5\,\text{IQR} = 120 - 75 = \mathbf{45} \text{ bpm}$$

$$\text{Abnormally high: } x > P_{75} + 1.5\,\text{IQR} = 170 + 75 = \mathbf{245} \text{ bpm}$$

Rule: flag any patient outside [45, 245] bpm. (This is the data-driven IQR/boxplot outlier rule; if $\bar{x}$ and s were given instead, the $\bar{x} \pm 3s$ rule would be the alternative.)

12.3 Q3 — Mean 250 vs. median 220 mg/dL [FROM SYLLABUS]

Mean > median $\Rightarrow$ the distribution is **positively (right) skewed**: a few patients with very high cholesterol (outliers) pull the mean upward, while the median stays at the typical patient. **Report the median** (220 mg/dL) — it is robust to outliers/skew and better represents the typical cholesterol level; the mean is distorted by extreme values.

12.4 Q4 — Draw the histogram (15 cholesterol values) [FROM SYLLABUS]

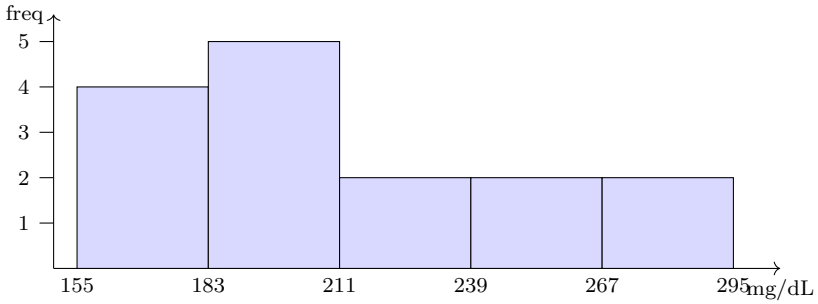
Data: $155, 160, 165, 170, 185, 190, 195, 200, 210, 220, 230, 250, 260, 280, 295$ ($n = 15$, already sorted).

Step 1: min $= 155$, max $= 295$, $R = 295 - 155 = 140$.
Step 2: $k = \lceil \log_2 15 + 1 \rceil = \lceil 3.91 + 1 \rceil = \lceil 4.91 \rceil = 5$ bins.

Step 3: $w = \lceil 140/5 \rceil = 28$.
Step 4: bins from min, include–exclude; $5 \times 28 = 140$ exactly, so close the last bin to keep 295:

Bin	Interval	Members	Freq
B_1	$[155, 183)$	155,160,165,170	4
B_2	$[183, 211)$	185,190,195,200,210	5
B_3	$[211, 239)$	220,230	2
B_4	$[239, 267)$	250,260	2
B_5	$[267, 295]$	280,295	2

(total = 15 ✓)



(If you instead choose round bins of width 30 starting at 150 — $[150, 180)$:4, $[180, 210)$:4, $[210, 240)$:3, $[240, 270)$:2, $[270, 300)$:2 — state the method; but the teacher's algorithm gives $w = 28$.)

12.5 Q5 — Two datasets: classification or regression? [FROM SYLLABUS]

Dataset-1 (cybersecurity):

(a) **Regression problem** — the output is a continuous numerical quantity (days).
(b) Target: `days_to_breach`. Features: `password_length`, `num_prev_phishing_clicks`, `department`, `2FA_enabled`, `last_password_change_days`.
(c) Numeric: `password_length`, `num_prev_phishing_clicks`, `last_password_change_days` (integers/-days). Categorical: `department` (nominal labels A, B, C), `2FA_enabled` (boolean 0/1 = binary categorical).
(d) Target type: **continuous numerical** variable.

Dataset-2 (email spam):

(a) **Classification problem** — the output is a category (spam / not spam).
(b) Target: `Spam`. Features: `Email_Text`, `Sender_Domain`, `Number_of_Links`, `Contains_Images`.
(c) Numeric: `Number_of_Links` (discrete integer). Categorical: `Sender_Domain` (nominal), `Contains_Images` (boolean), `Email_Text` (raw text = unstructured/textual data).
(d) Target type: **categorical (binary: 0/1)** variable.

12.6 Q6 — Define outlier, population, sample [FROM SYLLABUS]

Use the definitions boxed in §2 word-for-word.

13 SOLVED: 2021 semester paper (supporting practice)

13.1 Q2 — Bookstore waiting times: box-and-whisker, skew, outliers [FROM SYLLABUS]

Data ($n = 32$): 45, 17, 29, 28, 25, 45, 24, 26, 5, 24, 27, 29, 29, 35, 8, 53, 1, 23, 40, 1, 37, 25, 28, 28, 3, 30, 25, 60, 4, 29, 53, 26.

Sorted:

1	1	3	4	5	8	17	23
24	24	25	25	25	26	26	27
28	28	28	29	29	29	29	30
35	37	40	45	45	53	53	60

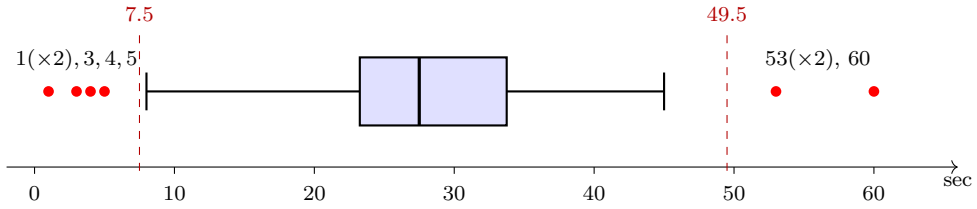
Quartiles (teacher's rank method, $r = \frac{k}{100}(n+1) = k\% \times 33$):

$$Q_1: \ r = 8.25 \Rightarrow x_8 + 0.25(x_9 - x_8) = 23 + 0.25(1) = 23.25$$

$$Q_2: \ r = 16.5 \Rightarrow \frac{x_{16}+x_{17}}{2} = \frac{27+28}{2} = 27.5 \qquad Q_3: \ r = 24.75 \Rightarrow 30 + 0.75(35-30) = 33.75$$

IQR $= 33.75 - 23.25 = 10.5$.

(i) Box plot. Mild fences: $23.25 - 1.5(10.5) = 7.5$ and $33.75 + 1.5(10.5) = 49.5$. Whiskers go to the extreme values *inside* the fences: 8 (left) and 45 (right).



Q_1=23.25, Q_2=27.5, Q_3=33.75; whiskers 8 and 45; red dots = outliers.

(ii) Skew: $Q_3 - Q_2 = 6.25 > Q_2 - Q_1 = 4.25$ — the median sits closer to Q_1 (upper half of the box is longer) $\Rightarrow$ mildly **positively (right) skewed**, with stragglers on *both* ends (very fast 1–5 s and very slow 53–60 s customers).

(iii) Mild / extreme outliers (rank method). Mild fences [7.5, 49.5]; extreme fences: $23.25 - 3(10.5) = -8.25$ and $33.75 + 3(10.5) = 65.25$. Outside mild fences: $\{1, 1, 3, 4, 5\}$ (low) and $\{53, 53, 60\}$ (high) $= 8$ values; none is outside the extreme fences $\Rightarrow$ **8 mild outliers, 0 extreme**.

(iv) Portion: mild $= 8/32 =$ **25**%; extreme $= 0/32 =$ **0**%.

Exam trap

Method sensitivity. Tukey's "median-of-halves" method gives $Q_1 = 23.5$, $Q_3 = 32.5$, IQR $= 9$, fences [10, 46] $\Rightarrow$ 9 mild outliers ($1, 1, 3, 4, 5, 8$ low; $53, 53, 60$ high). Different methods shift the numbers slightly — in the exam *state your method* (default here: teacher's rank method $r = \frac{k}{100}(n+1)$) and stay consistent throughout.

13.2 Q3 — Rainfall vs. runoff: fit line, predict, correlate [FROM SYLLABUS]

$n = 15$. Build the sums first (show this table — it earns method marks):

$$\Sigma x = 798, \quad \Sigma y = 643, \quad \Sigma x^2 = 63040, \quad \Sigma y^2 = 41999, \quad \Sigma xy = 51232$$

$$S_{xx} = \Sigma x^2 - \frac{(\Sigma x)^2}{n} = 63040 - 42453.6 = 20586.4 \qquad S_{yy} = 41999 - 27563.27 = 14435.73$$

$$S_{xy} = \Sigma xy - \frac{(\Sigma x)(\Sigma y)}{n} = 51232 - 34207.6 = 17024.4 \qquad \bar{x} = 53.2, \quad \bar{y} = 42.87$$

(a) Regression line:

$$w = \frac{S_{xy}}{S_{xx}} = \frac{17024.4}{20586.4} = 0.827 \qquad b = \bar{y} - w\bar{x} = 42.87 - 0.827(53.2) = -1.13$$

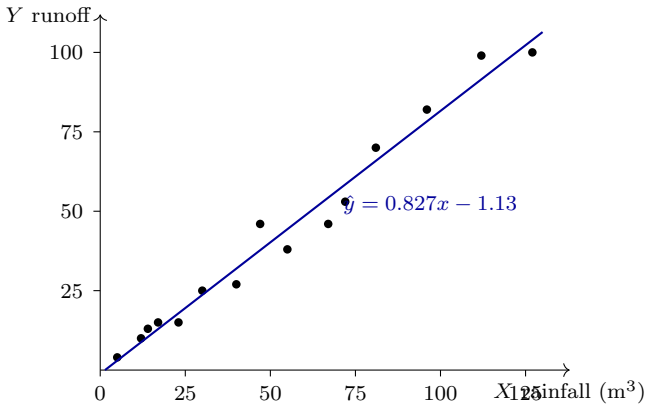
$$\boxed{\hat{y} = 0.827\,x - 1.13} \quad \text{(runoff} \approx 0.83 \times \text{rainfall)}$$

(b) Prediction at $x = 50$: $\hat{y} = 0.827(50) - 1.13 =$ **40.2 m^3**.

(c) Correlation & determination:

$$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} = \frac{17024.4}{\sqrt{20586.4 \times 14435.73}} = \frac{17024.4}{17238.9} = \mathbf{0.988} \qquad r^2 = \mathbf{0.975}$$

Comment: $r = 0.988 \Rightarrow$ near-perfect **strong positive linear** relationship; $r^2 = 0.975 \Rightarrow 97.5\%$ of the variation in runoff is explained by rainfall — the linear fit is excellent.



13.3 Q1 — Wind-speed frequency table: mode, kurtosis [PARTIALLY IN SYLLABUS]

(i) Mode (grouped data). Modal class = highest frequency 209 $\Rightarrow$ [1.78, 2.67), $L = 1.78$, $f_1 = 209$, $f_0 = 128$, $f_2 = 106$, $h = 0.89$:

$$\text{Mode} = L + \frac{f_1 - f_0}{2f_1 - f_0 - f_2}\,h = 1.78 + \frac{81}{418 - 234}(0.89) = 1.78 + 0.39 = \mathbf{2.17\ knots}$$

(Useful anchors: $N = \Sigma f = 654$; grouped mean via midpoints $\bar{x} = \frac{\Sigma f x_{mid}}{N} \approx 2.58$.)

(ii) Karl Pearson's coefficient of kurtosis [OFF SYLLABUS] — formula only (not in class notes):

$$\beta_2 = \frac{m_4}{m_2^2}, \qquad m_r = \frac{1}{N}\sum f\,(x_{mid} - \bar{x})^r$$

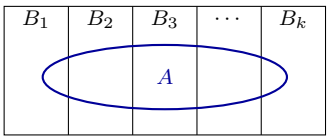
Build columns x_{mid}, f, $d = x_{mid} - \bar{x}$, fd^2, fd^4; then β_2. **(iii) Comment rule:** $\beta_2 > 3$ leptokurtic (sharper peak than normal), $\beta_2 = 3$ mesokurtic, $\beta_2 < 3$ platykurtic (flatter). Low priority for tomorrow.

14 JU probability CT — [OFF SYLLABUS] rapid answers (skim only)

Nothing below appears in Stat-01–07; keep for safety, 10 minutes max.

1(a) **Experiment:** a repeatable process with well-defined outcomes whose result cannot be predicted with certainty. **Sample space S:** the set of *all* possible outcomes.

1(b) Total probability with partition $B_1, \ldots, B_k$ of S: $P(A) = \sum_{i=1}^{k} P(A \cap B_i) = \sum_{i=1}^{k} P(B_i)\,P(A \mid B_i)$.



1(c) **Chebyshev's inequality:** for any $k > 0$, $P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$ (equivalently $P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}$).

1(d) $H(X) = 1 \iff x^2 = 1 \iff x = \pm 1$ (both in $[-1, 1]$): $P\big(H(X) = 1\big) = P(X = -1) + P(X = 1) = \frac{1}{4} + \frac{2}{5} = \boxed{\frac{13}{20}}$. (Given masses sum to $\frac{19}{20}$; the leftover $\frac{1}{20}$ sits on other x, where $H = -1$ — irrelevant.)

1(e) **Method:** at every junction each forward edge has probability $\frac{1}{\#\text{choices}}$; multiply along a path, add over all $A \to B$ paths. For the symmetric diamond shown (2 choices at A, 2 at each middle node, both lower nodes feeding B): 4 paths $\times \frac{1}{2} \cdot \frac{1}{2} = \ P(B) = 1$. If your figure version has a dead-end/bypass edge, delete those paths and re-add — the enumeration technique is what carries the marks.

2(a) $F_X(x) = 1 - (\frac{1}{2})^{x+1}$, $x = 0, 1, 2, \ldots$

(i) pmf: $p(x) = F(x) - F(x-1) = (\frac{1}{2})^x - (\frac{1}{2})^{x+1} = \boxed{(\frac{1}{2})^{x+1}}$ (check: $\sum_{x \geq 0} (\frac{1}{2})^{x+1} = 1$ ✓).

(ii) $P(0 \leq X \leq 8) = F(8) = 1 - (\frac{1}{2})^9 = \boxed{\frac{511}{512} \approx 0.998}$.

2(b) $Y = 2000(12 - X)$:

x	10	11	12	13	14
p	0.1	0.3	0.4	0.1	0.1
y	4000	2000	0	-2000	-4000

$E(X) = 1 + 3.3 + 4.8 + 1.3 + 1.4 = \mathbf{11.8}$; $E(X^2) = 140.4 \Rightarrow V(X) = 140.4 - 11.8^2 = \mathbf{1.16}$.
$E(Y) = 2000\,(12 - E(X)) = 2000(0.2) = \mathbf{400}$; $V(Y) = 2000^2\, V(X) = \mathbf{4{,}640{,}000}$.

15 Exam-hall page (read 30 min before)

Center
$\bar{x} = \frac{1}{n}\sum x_i$; median: sort, $mid = \frac{n+1}{2}$;
mode = most frequent (categorical).
Outlier? ⇒ median. Categorical? ⇒ mode.
mean > median ⇒ right/positive skew.

Spread
$R = \max - \min$; $s^2 = \frac{1}{n-1}\sum(x_i - \bar{x})^2$;
$\sigma^2 = \frac{1}{N}\sum(x_i - \mu)^2$; $s = \sqrt{s^2}$ (units!).

3σ rule
outlier if $x < \bar{x} - 3s$ or $x > \bar{x} + 3s$; else normal.

Percentile (rank method)
sort; $r = \frac{k}{100}(n+1)$; integer ⇒ $P_k = x_r$;
else $P_k = x_{\lfloor r \rfloor} + d\,(x_{\lfloor r \rfloor + 1} - x_{\lfloor r \rfloor})$, $d = r - \lfloor r \rfloor$.
Reverse: $k = \frac{r}{n} \times 100$.

IQR / box plot
$\text{IQR} = Q_3 - Q_1$; $\text{LB} = Q_1 - 1.5\,\text{IQR}$;
$\text{UB} = Q_3 + 1.5\,\text{IQR}$; outside ⇒ outlier.
Extreme: beyond $Q_1 - 3\text{IQR}$ / $Q_3 + 3\text{IQR}$.
5-number: min, Q_1, Q_2, Q_3, max.
Whisker ends at farthest value *inside* fence.

Histogram (Sturges)
$R = \max - \min$; $k = \lceil \log_2 n + 1 \rceil$; $w = \lceil R/k \rceil$;
bins $[a, a+w)$ include-exclude, start at min.
$\log_2$: 8→3, 16→4, 32→5, 15→3.91.
Bar chart = categorical (gaps); histogram = continuous.

Cov / correlation
$\text{Cov} = \frac{1}{n-1}\sum(x_i - \bar{x})(y_i - \bar{y})$;
$r = \frac{\text{Cov}}{s_x s_y} \in [-1, 1]$; $r = \cos\theta$; 0.7–0.9 strong.

Regression
$\hat{y} = wx + b$; $C(w,b) = \frac{1}{N}\sum(y_i - wx_i - b)^2 \to \min$;
$w = S_{xy}/S_{xx}$; $b = \bar{y} - w\bar{x}$; r^2 = % explained.

ML decision (Q5 type)
target continuous ⇒ regression;
target categorical ⇒ classification;
boundary: line (2D) / plane (3D) / hyperplane.
Supervised = labelled; unsupervised = unlabelled;
RL = agent, environment, action, reward/penalty.

Definitions (2 free marks)
Outlier / population (N, μ) / sample $(n, \bar{x})$ — §2 box.

Last-minute traps (each one has cost real marks)

1. **Sort first.** Every median/percentile/quartile answer is wrong if you skip sorting.
2. Rank uses $(n{+}1)$, not n: $r = \frac{k}{100}(n+1)$.
3. Never report the mean when an outlier exists — say "median, because robust to outliers."
4. Whiskers stop at the farthest data value *within* the fences (27, 45…), never at LB/UB.
5. Bin edges: include left, exclude right; if max lands on the last edge, close that bin.
6. Sample variance divides by $n-1$ (DOF); population by N. Variance units are squared.
7. r measures only *linear* relation; $r = 0 \neq$ "no relation at all."
8. Always *interpret* after computing (consistency, skew, outlier meaning) — interpretation lines are where the "Explain" marks live.

Night-before checklist

☐ Redo 32-CT Q1 and Q4 on paper, timed (18 min total). ☐ Recompute Q_1, Q_3, fences for $\{22, \ldots, 60\}$ from memory. ☐ Recite the three definitions. ☐ Write Sturges' two formulas once. ☐ Sleep.